

NONPARAMETRIC ELIMINATION SELECTION PROCEDURES BASED ON ROBUST ESTIMATORS

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SUMMARY

Let $\Pi_1, \Pi_2, \dots, \Pi_k$ be k ($k \geq 2$) independent populations and assume that the distribution function of Π_i depends on some (unknown) parameter(s) μ_i ($i = 1, \dots, k$). This paper is concerned with the problem of constructing sequential elimination procedures for selecting the population having maximum μ_i , which are in some sense nonparametric and robust. Except for some regularity assumptions, no assumptions are made about the form of the distribution function F_i of Π_i , unlike the case in e.g. Robbins, Sobel and Starr (1968) and Swanepoel and Geertsema (1973) where it is assumed that $F_i = F(x - \mu_i)$ with F symmetric about 0. However, a special case of our results is to take $F_i = F((x - \mu_i)/\sigma_i)$, yielding the first elimination procedures for selecting the population with the largest location parameter when the scale parameters are unequal. Using the indifference zone formulation, procedures are proposed and studied which are based on certain robust estimators such as M-estimators and L-estimators. Asymptotic expressions are obtained for the expected sample sizes of the procedures which show that, asymptotically, our procedures are more efficient than existing procedures. Finally, a class of elimination selection procedures, which are asymptotically nonparametric, are given for the subset approach of Gupta (1965). These procedures have the property of choosing, with large probability, only one population in the subset when sampling is terminated.

1. Introduction

Let $\Pi_1, \Pi_2, \dots, \Pi_k$ be k ($k \geq 2$) independent populations and assume that the distribution function of Π_i depends on some (unknown) parameter(s) μ_i ($i = 1, \dots, k$). This paper is concerned with the problem of constructing sequential elimination procedures for selection of the population having maximum μ_i , which are in some sense nonparametric and robust.

Paulson (1964) considers an elimination procedure (using the indifference zone formulation) for selecting the normal population with the largest mean, all populations

having a common variance. Swanepoel and Venter (1975) give a method for the construction of elimination selection procedures in the normal case.

Also, a general method for calculating the asymptotic expected number of observations as the probability of correct selection tends to one, is presented. They propose a class of procedures and the best ones in the above asymptotic sense are found, showing that Paulson's procedure is not one of these. All these procedures are based on the sample mean. However, the sample mean is notoriously non-robust, which, in this context, implies that too many observations are being taken in most cases. In order to reduce the number of observations and to obtain desired robustness properties, a class of elimination procedures are proposed and studied in section 3, which solve the problem by using certain robust estimators such as M-estimators and L-estimators. In section 4 a class of elimination selection procedures, which are asymptotically nonparametric, are given for the subset selection approach. These procedures are based on a general class of one-sample rank order statistics. Since the method for solving these problems is based on the theory of weak convergence, two invariance theorems are stated and proved in section 2.

2. Two invariance theorems

Let $\{X_n, n \geq 1\}$ be a sequence of independent, identically distributed random variables, $M_n = M_n(X_1, X_2, \dots, X_n)$ and $R_n = R_n(X_1, X_2, \dots, X_n)$ two statistics satisfying

$$(1) \quad M_n - \theta = n^{-1} \sum_{i=1}^n Y_i + R_n \quad \text{a.s. as } n \rightarrow \infty$$

where Y_i is a function of X_i only and θ is some constant.

$$(2) \quad n^{1/2} R_n \rightarrow 0 \quad \text{a.s. as } n \rightarrow \infty,$$

where $Y_1, Y_2, \dots$ are i.i.d. random variables such that $E(Y_1) = 0$, $E(Y_1^2) = \sigma^2$ and $E\{Y_1^2(\log_2 Y_1^2)\} < \infty$. Let $V_n^2 = V_n^2(X_1, X_2, \dots, X_n)$ be a strongly consistent estimator of σ^2 such that

$$(3) \quad V_n^2 = \sigma^2 + o[(\log_2 n)^{-1/2}] \quad \text{a.s. as } n \rightarrow \infty.$$

Two invariance theorems will now be proved which can be considered as extensions of the invariance theorems of Robbins and Siegmund (1970).

Theorem 2.1

Suppose that $g(t)$ is a nonnegative continuous function for $t > 0$ satisfying

$$(i) \quad t^{-1/2} g(t) \uparrow \infty \text{ as } t \downarrow 0 \text{ and } t^{-1/2} g(t) \uparrow \infty \text{ as } t \uparrow \infty \text{ and}$$

$$(ii) \int_{0+}^{\infty} t^{-3/2} g(t) \exp \{-g^2(t)/8t\} dt < \infty.$$

Then

$$\lim_{m \rightarrow \infty} P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } n \geq 1)$$

$$= P(W(t) \geq g(t) \text{ for some } t > 0),$$

where $W(t)$ is a standard Wiener process for $0 \leq t < \infty$.

Proof

For any $0 < a < c < \infty$ we have

$$(4) \quad P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } am \leq n \leq cm)$$

$$\leq P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } n \geq 1)$$

$$\leq P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } 1 \leq n < am)$$

$$+ P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } am \leq n \leq cm)$$

$$+ P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } n > cm)$$

Let $[x]$ denote the largest integer less than or equal to x . From (1) and Donsker's invariance principle it follows, upon application of (for example) theorem 4.1 of Billingsley (1968), that the process $W_m(t) = \{\sigma^{-1} m^{-1/2} [mt](M_{[mt]} - \theta) - g([mt]/m)\}$ weakly converges to $\{W(t) - g(t)\}$. Hence, for any $0 < a < c < \infty$, $\sup \{W_m(t): a \leq t \leq c\}$ converges in law to $\sup \{W(t) - g(t): a \leq t \leq c\}$, the distribution of which has 0 as a continuity point — see lemma 3 of Robbins and Siegmund (1970). This implies

$$(5) \quad \lim_{m \rightarrow \infty} P(\sigma^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } am \leq n \leq cm)$$

$$= P(W(t) \geq g(t) \text{ for some } a \leq t \leq c).$$

We now prove that

$$(6) \quad \lim_{m \rightarrow \infty} P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } am \leq n \leq cm) = \lim_{m \rightarrow \infty} P_1 \text{ (say)}$$

$$= P(W(t) \geq g(t) \text{ for some } a \leq t \leq c).$$

Let $\epsilon > 0$ be arbitrary. Using (5) we find

$$\begin{aligned}
 P_1 &= P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } am \leq n \leq cm, V_n > \sigma - \epsilon \\
 &\quad \text{for all } n \geq am) \\
 &\quad + P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } am \leq n \leq cm, V_n \leq \sigma - \epsilon \\
 &\quad \text{for some } n \geq am) \\
 &\leq P(\sigma^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)(\sigma - \epsilon)/n\sigma \text{ for some } am \leq n \leq cm) \\
 &\quad + P(V_n \leq \sigma - \epsilon \text{ for some } n \geq am) \\
 &\rightarrow P(W(t) \geq g(t)(\sigma - \epsilon)/\sigma \text{ for some } a \leq t \leq c) \text{ as } m \rightarrow \infty \\
 &\rightarrow P(W(t) \geq g(t) \text{ for some } a \leq t \leq c) \text{ as } \epsilon \rightarrow 0.
 \end{aligned}$$

A similar argument proves the reverse inequality.

Furthermore, using (1) we get

$$\begin{aligned}
 (7) \quad &P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } 1 \leq n < am) \\
 &\leq P(V_n^{-1} \sum_{i=1}^n Y_i - W(n) \geq m^{1/2} g(n/m)/4 \text{ for some } 1 \leq n < am) \\
 &\quad + P(V_n^{-1} nR_n \geq m^{1/2} g(n/m)/4 \text{ for some } 1 \leq n < am) \\
 &\quad + P(W(n) \geq m^{1/2} g(n/m)/2 \text{ for some } 1 \leq n < am) \\
 &= P_2 + P_3 + P_4 \text{ (say)}.
 \end{aligned}$$

From theorem 4.2 of Jain, Jogdeo and Stout (1975) it follows that

$$(8) \quad \sigma^{-1} \sum_{i=1}^n Y_i - W(n) = o(n^{1/2}) \text{ a.s. as } n \rightarrow \infty.$$

Applying (3), (i), (8) and the law of the iterated logarithm we obtain

$$\begin{aligned}
 (9) \quad &P_2 \leq P((V_n^{-1} \sum_{i=1}^n Y_i - W(n))/n^{1/2} \geq a^{-1/2} g(a)/4 \text{ for some } 1 \leq n < am) \\
 &\leq P((\sigma^{-1} \sum_{i=1}^n Y_i - W(n))/n^{1/2} \geq a^{-1/2} g(a)/8 \text{ for some } n \geq 1) \\
 &\quad + P((V_n^{-1} - \sigma^{-1}) \sum_{i=1}^n Y_i / n^{1/2} \geq a^{-1/2} g(a)/8 \text{ for some } n \geq 1)
 \end{aligned}$$

$$\rightarrow 0 \text{ as } a \rightarrow 0.$$

Also,

$$(10) P_3 \leq P(V_n^{-1} n^{1/2} R_n \geq a^{-1/2} g(a)/4 \text{ for some } n \geq 1)$$

$\rightarrow 0$ as $a \rightarrow 0$ using (2) and (i).

Using the fact that $\{\alpha^{-1/2} W(\alpha t)\}$ is a standard Wiener process and applying Kolmogorov's test (see Itô and Mc Kean (1965), p. 34) we get

$$(11) P_4 \leq P(W(t) \geq m^{1/2} g(t/m)/2 \text{ for some } 0 < t < am)$$

$$= P(W(t) \geq g(t)/2 \text{ for some } 0 < t < a)$$

$\rightarrow 0$ as $a \rightarrow 0$.

It therefore follows from (7), (9), (10) and (11) that

$$(12) P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } 1 \leq n < am) \rightarrow 0 \text{ as } a \rightarrow 0.$$

In exactly the same way as (12) it also follows that

$$(13) P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } n > cm) \rightarrow 0 \text{ as } c \rightarrow \infty.$$

The proof of the theorem now follows from (4), (6), (12) and (13) by first letting $m \rightarrow \infty$ and then letting $a \rightarrow 0$, $c \rightarrow \infty$.

Theorem 2.2

Let $g(t)$ be a nonnegative continuous function for $0 < t \leq 1$ such that

$$(i) \quad t^{-1/2} g(t) \uparrow \infty \text{ as } t \downarrow 0$$

$$(ii) \quad \int_{0+}^1 t^{-3/2} g(t) \exp \{-g^2(t)/8t\} dt < \infty.$$

Then

$$\lim_{m \rightarrow \infty} P(V_n^{-1} (M_n - \theta) \geq m^{1/2} g(n/m)/n \text{ for some } 1 \leq n \leq m)$$

$$= P(W(t) \geq g(t) \text{ for some } 0 < t \leq 1).$$

Proof

The proof is similar to that of theorem 2.1 and is omitted.

Remarks

- (a) Theorems 2.1 and 2.2 continue to hold if $M_n - \theta$ and $W(t)$ are replaced by $|M_n - \theta|$ and $|W(t)|$.
- (b) The proofs of the above theorems seem to be easier than those of Robbins and Siegmund (1970), p. 1420.

3. Elimination selection procedures – indifference zone approach

Let $\Pi_1, \Pi_2, \dots, \Pi_k$ denote k populations ($k \geq 2$) and let X_{ij} denote the j -th observation on population Π_i (for $i = 1, \dots, k$ and $j = 1, 2, \dots$), all such observations being assumed independent random variables. Assume the distribution function of Π_i is unknown and depends on some (unknown) parameter μ_i ($i = 1, \dots, k$). Assume, without loss of generality, that the indexing is such that $\mu_1 \leq \mu_2 \leq \dots \leq \mu_k$. If $\mu_k > \mu_{k-1}$, we refer to Π_k as the best population. Let

$$(14) \quad \Omega(\Delta^*) = \{\underline{\mu} = (\mu_1, \dots, \mu_k) : \mu_k - \mu_{k-1} \geq \Delta^*\}.$$

We wish to construct elimination procedures having the property

$$(15) \quad \liminf_{\Delta^* \rightarrow 0} P(\text{CS} / \underline{\mu} \in \Omega(\Delta^*)) \geq P^*$$

where CS denotes the event of a correct selection and P^* and Δ^* are given numbers with $1/k < P^* < 1$ and $\Delta^* > 0$.

Let $M_{in} = M_{in}(X_{i1}, X_{i2}, \dots, X_{in})$ be a strongly consistent estimator of μ_i ($i = 1, \dots, k$) such that

$$(16) \quad M_{in} - \mu_i = n^{-1} \sum_{j=1}^n Y_{ij} + R_{in} \quad \text{a.s. as } n \rightarrow \infty,$$

$$(17) \quad n^{1/2} R_{in} \rightarrow 0 \quad \text{a.s. as } n \rightarrow \infty,$$

where $Y_{i1}, Y_{i2}, \dots, Y_{in}$ are i.i.d. random variables satisfying

$$(18) \quad E(Y_{i1}) = 0, E(Y_{i1}^2) = \sigma_i^2 \text{ and } E\{Y_{i1}^2 (\log_2 Y_{i1}^2)\} < \infty.$$

Let $V_{in}^2 = V_{in}^2(X_{i1}, \dots, X_{in})$ be a strongly consistent estimator of σ_i^2 such that

$$(19) \quad V_{in}^2 = \sigma_i^2 + o[(\log_2 n)^{-1/2}] \text{ a.s. as } n \rightarrow \infty.$$

Suppose that $g(t, P^\star)$ is a function of t and $P^\star$ and for simplicity we write $g(t) \approx g(t, P^\star)$.

Throughout this section it is assumed that $g(t)$ satisfies the assumptions of theorem 2.1.

Let $r = r(\Delta^\star)$ be the largest integer less than or equal to $1/\Delta^\star$. Then our elimination procedure is defined as follows:

Begin at stage one with a single observation from each of the k populations and compute the statistics M_{i1}, V_{i1}^2 for $i = 1, \dots, k$. Suppose that $M_{s1} = \max \{M_{11}, \dots, M_{k1}\}$. If for some $i \neq s$

$$M_{i1} \leq M_{s1} - r^{1/2} g(1/r) [V_{i1}^2 + V_{s1}^2]^{1/2} + \Delta^\star$$

then Π_i is eliminated from further consideration. If only one population remains, it is declared best and if more than one remains, we proceed to the next stage by taking a single observation from each of the remaining populations. At the n -th stage, a single observation is made from each of the remaining populations, and the statistics M_{in}, V_{in}^2 are computed for all remaining populations. Let $M_{jn} = \max_{\alpha} M_{\alpha n}$, where the maximum is taken over all populations which have not been eliminated after the $(n-1)$ -th stage. If for some $i \neq j$

$$(20) \quad M_{in} \leq M_{jn} - r^{1/2} g(n/r) [V_{in}^2 + V_{jn}^2]^{1/2}/n + \Delta^\star,$$

then Π_i is eliminated and the procedure repeated.

Theorem 3.1

Let the function $g(t)$ be defined by

$$(21) \quad P(W(t) \geq g(t) \text{ for some } t > 0) = (1 - P^\star)/(k-1).$$

If the procedure stops with probability one, then

$$\liminf_{\Delta^\star \rightarrow 0} P(CS/\mu \in \Omega(\Delta^\star)) \geq P^\star.$$

Proof

For all $\mu \in \Omega(\Delta^\star)$ we get

$1 - P(\text{CS}) \leq P(\text{For at least one value of } j (j = 1, \dots, k-1))$

$$\begin{aligned}
 & M_{kn} \leq M_{jn} - r^{1/2} g(n/r) [V_{kn}^2 + V_{jn}^2]^{1/2} / n + \Delta^* \text{ for some } n \geq 1) \\
 & \leq \sum_{j=1}^{k-1} P(M_{kn} \leq M_{jn} - r^{1/2} g(n/r) [V_{kn}^2 + V_{jn}^2]^{1/2} / n + \Delta^* \text{ for some } n \geq 1) \\
 & \leq \sum_{j=1}^{k-1} P(M_{jn} - M_{kn} - (\mu_j - \mu_k) \geq r^{1/2} g(n/r) [V_{jn}^2 + V_{kn}^2]^{1/2} / n \text{ for some } n \geq 1) \\
 & \rightarrow P(W(t) \geq g(t) \text{ for some } t > 0)(k-1) \\
 & = 1 - P^*,
 \end{aligned}$$

which follows from theorem 2.1, (16), (17), (18) and (19).

We now consider one possible choice of $g(t)$. Define

$$(22) \quad g(t) = (t + 1/4)^{1/2} \alpha^{-1}(\log(t + 1/4) + 2\log((k-1)/2(1 - P^*)))$$

where $\alpha^{-1}(x)$ is the inverse function of $\alpha(x) = x^2 + 2\log\Phi(x)$ with Φ the standard normal cdf. From Robbins and Siegmund (1970, p. 1413) it follows that $g(t)$ satisfies (21) and the assumptions of theorem 2.1. An explicit expression for $g(t)$ is not possible, but sharp bounds are

$$\begin{aligned}
 (23) \quad & (t + 1/4)^{1/2} [\log(t + 1/4) + 2\log((k-1)/2(1 - P^*))]^{1/2} \\
 & \leq g(t) \leq (t + 1/4)^{1/2} [\log(t + 1/4) + 2\log((k-1)/(1 - P^*))]^{1/2}.
 \end{aligned}$$

If we replace the procedure in (20) by one in which $g(t)$ is taken equal to the upper bound above, theorem 3.1 would still hold. (For other choices of $g(t)$ satisfying (21), see Robbins and Siegmund (1970, p. 1413).)

Let T denote the total number of observations required to stop, using (for the rest of this section) $g(t)$ defined by (22). Then we have

Theorem 3.2

$P(T < \infty) = 1$ for all configurations of the parameters $\mu_1, \dots, \mu_k$.

Proof

Define the following auxiliary stopping variable: Consider the case where at each stage of sampling an independent observation is obtained from *each* of the k population and let M be the first n such that

$$r^{1/2} g(n/r) \sum_{i=1}^k \sum_{j=1}^k [V_{in}^2 + V_{jn}^2]^{1/2} / n < \Delta^*.$$

Since for each fixed $\Delta^* > 0$ $g(n/r)/n \rightarrow 0$ as $n \rightarrow \infty$ and $V_{sn}^2 \rightarrow \sigma_s^2 > 0$ a.s. ($s = 1, \dots, k$) it follows that $M < \infty$ a.s. Also, it can easily be shown that $T \leq kM$.

Remark

One possible choice of M_{in} can be the trimmed mean. De Wet and Venter (1974) have shown that (16), (17), (18) and (19) hold for M_{in} . For more general L-estimators and certain M-estimators Carroll (1975) has given similar asymptotic representations.

We shall now develop asymptotic theory as $P^* \rightarrow 1$ which should be of importance since in practice values of P^* close to one are usually chosen. (See for instance Bechhofer, Kiefer and Sobel (1968) ch. 6 where extensive asymptotic theory as $P^* \rightarrow 1$ is developed.) Asymptotic expressions as $\Delta^* \rightarrow 0$ for expected sample sizes of elimination procedures are much harder to obtain and it seems that no work has yet been devoted to it in the literature.

Theorem 3.3

If $\mu_k > \mu_{k-1}$, then

$$\lim_{P^* \rightarrow 1} E(T) / (-2 \log(1 - P^*)) = \sum_{i=1}^{k-1} y_{(i)} + y_{(k-1)},$$

where $y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(k-1)}$ are the values of $y_1, \dots, y_{k-1}$ arranged in increasing order, where

$$y_i = (\sigma_k^2 + \sigma_i^2) / (\mu_k - \mu_i + \Delta^*)^2 \text{ for } i = 1, \dots, k-1.$$

Proof

The proof is similar to the proof of theorem 3 of Swanepoel and Geertsema (1976) and is omitted.

Asymptotic efficiencies

In Swanepoel and Geertsema (1976) (see also Swanepoel and Venter (1975)) an elimination procedure is given for the problem of selecting the *normal* population with the largest mean, all populations having a common known variance σ^2 . (Let T_S denote the total number of observations required to stop by this procedure.) They prove that their procedure is asymptotically as $P^* \rightarrow 1$ (in terms of expected sample sizes) more efficient than those of Paulson (1964) and Bechhofer, Kiefer and Sobel (1968, p. 161). From theorem 3.3 and theorem 3 of Swanepoel and Venter (1975, see top of p. 114) it

follows that

$$(24) \lim_{P^* \rightarrow 1} E(T)/E(T_S) = 1$$

where T is the total number of observations taken by the procedure of this section. (M_{in} is taken to be the sample mean and we assume $\sigma_i^2 = \sigma^2$ for all i .) Recall that this procedure requires no knowledge of the underlying distribution function F_i of Π_i . Hence, from (24) one can conclude that this procedure is "asymptotically efficient" in the sense of Chow and Robbins (1965).

A procedure with which our procedure may be compared, is one by Robbins, Sobel and Starr (1968). We shall denote the total number of observations needed by this procedure by T_R . This procedure does not require knowledge of an assumed common variance σ^2 . However, our procedure does not even assume a common variance and it also has the desirable elimination property. We now consider the relative efficiency of these two procedures as $P^* \rightarrow 1$ and in the least-favourable configuration where $\mu_1 = \dots = \mu_{k-1} = \mu_k - \Delta^*$. We point out that the asymptotic expected sample size of our procedure has its maximum (outside the indifference zone) in this configuration, while the asymptotic expression for $E(T_R)$ does not depend on the specific configuration. In fact, it may be shown that

$$(25) \lim_{P^* \rightarrow 1} E(T_R)/(-4\log(1 - P^*)) = k\sigma^2/(\Delta^*)^2.$$

From theorem 3.3 it follows that

$$(26) \lim_{P^* \rightarrow 1} E(T)/(-2\log(1 - P^*)) = k\sigma^2/2(\Delta^*)^2.$$

(M_{in} is taken to be the sample mean and we assume $\sigma_i^2 = \sigma^2$ for all i .)

From (25) and (26) we obtain

$$(27) \lim_{P^* \rightarrow 1} E(T)/E(T_R) = 1/4.$$

It may be noted that this efficiency may be considerably less than 1/4 if, in the absence of normality, we base our procedure on some suitable robust estimator. (The procedure of Robbins, Sobel and Starr (1968) is designed only for the case where the estimator of μ_i is the sample mean.)

4. Elimination selection procedures – subset approach

Let $\Pi_1, \Pi_2, \dots, \Pi_k$ denote k populations, the distribution function of Π_i being

$F(x - \theta_i)$ where F is an unknown absolutely continuous distribution function and θ_i is an unknown location parameter for Π_i . Using the subset approach of Gupta (1965), this section is concerned with the problem of constructing sequential elimination procedures for selecting a subset of the populations which includes the best population. Let CS denote the event of a correct selection, i.e. the event that the best population is contained in the selected subset.

More specifically, if the indexing is such that $\theta_1 \leq \theta_2 \leq \dots \leq \theta_k$, then we wish to construct procedures which guarantee the probability of correct selection for *all* configurations of the parameters θ_i ($i = 1, \dots, k$), unlike the case with the indifference zone approach where it is only guaranteed outside an indifference zone.

Let $X_{i\alpha}$ denote the α -th observation from Π_i (all such observations being assumed independent random variables); put

$$(28) \quad Y_{ij\alpha} = X_{i\alpha} - X_{j\alpha}, \quad \tau_{ij} = \theta_i - \theta_j$$

and define

$$(29) \quad c(u) = 1 \text{ for } u \geq 0, \quad c(u) = 0 \text{ for } u < 0.$$

Next let

$$(30) \quad R_{ij\alpha n}(\tau_{ij}) = \sum_{\beta=1}^n c(|Y_{ij\alpha} - \tau_{ij}| - |Y_{ij\beta} - \tau_{ij}|),$$

$$(31) \quad Z_{ijn}(\tau_{ij}) = n^{-1} \sum_{\alpha=1}^n c(Y_{ij\alpha} - \tau_{ij}) J_n(R_{ij\alpha n}(\tau_{ij})/(n+1)),$$

$$(32) \quad \bar{J}_n = n^{-1} \sum_{\alpha=1}^n J_n(\alpha/(n+1)), \quad A^2 = \int_0^1 J^2(u) du,$$

where $J_n(u)$, $J(u)$ ($0 < u < 1$) are defined as in Sen and Ghosh (1971), p. 190. Let $G(x)$ be the distribution function of $Y_{ij\alpha} - \tau_{ij}$, having density $g(x)$ w.r.t. Lebesgue measure, and assume

$$(33) \quad \lim_{x \rightarrow \infty} g(x)(d/dx)J[2G(x) - 1] < \infty.$$

Suppose that $g(t, P^*)$ is a function of t and P^* and for simplicity we write $g(t) = g(t, P^*)$. Throughout this section it is assumed that $g(t)$ satisfies the assumptions of theorem 2.2.

We now define our ranks procedure which is truncated at a point m , some positive constant.

Begin at stage one with a single observation from each of the k populations and compute the $k(k-1)$ statistics $Z_{ij1}(0)$. If for some $i \neq j$

$$Z_{ij1}(0) - \frac{1}{2}\bar{J}_1 \geq m^{1/2}g(1/m)A/2$$

then Π_j is eliminated from further consideration. If only one population remains, it is declared best and if more than one remains, we proceed to the next stage by taking a single observation from each of the remaining populations. At the n -th stage, $1 \leq n \leq m-1$, a single observation is made from each of the remaining populations, and the statistics $Z_{ijn}(0)$ are computed for all remaining populations. If for some $i \neq j$

$$(34) \quad Z_{ijn}(0) - \frac{1}{2}\bar{J}_n \geq m^{1/2}g(n/m)A/2n$$

then Π_j is eliminated and the procedure repeated. However, if the procedure reaches stage m (the truncation point) an observation is taken from each of the remaining populations and the elimination rule (34) is repeated; after this, we stop sampling and choose as subset all the remaining populations.

Theorem 4.1

Let the function $g(t)$ be defined by

$$(35) \quad P(W(t) \geq g(t) \text{ for some } 0 < t \leq 1) = (1 - P^*)/(k-1)$$

Then

$$\liminf_{m \rightarrow \infty} P(CS) \geq P^* \text{ for all configurations of the parameters } \theta_1, \theta_2, \dots, \theta_k$$

Proof

Note that $Y_{ij\alpha} - \tau_{ij}$ is symmetrically distributed about 0. Under the condition (33), Sen and Ghosh (1974), p. 550 have shown that

$$(36) \quad Z_{ijn}(\tau_{ij}) - \frac{1}{2}\bar{J}_n = n^{-1} \sum_{i=1}^n Y_i^* + R_n \text{ a.s. as } n \rightarrow \infty$$

with

$$n^{1/2}R_n \rightarrow 0 \text{ a.s. as } n \rightarrow \infty$$

where $Y_1^*, Y_2^*, \dots$ are i.i.d. random variables such that $E(Y_1^*) = 0$ and $E\{(Y_1^*)^2\} = A^2/4$. It is readily seen, that $Z_{ijn}(\tau_{ij})$ is non-increasing in τ_{ij} for all i and j ; hence

$$(37) \quad Z_{ikn}(\tau_{ik}) \geq Z_{ikn}(0).$$

From theorem 2.2, (36) and (37) we now get

$$\begin{aligned} 1 - P(\text{CS}) &\leq \sum_{i=1}^{k-1} P(\Pi_i \text{ eliminates } \Pi_k) \\ &\leq \sum_{i=1}^{k-1} P(Z_{ikn}(0) - \frac{1}{2} \bar{J}_n \geq m^{1/2} g(n/m) A/2n \text{ for some } 1 \leq n \leq m) \\ &\leq \sum_{i=1}^{k-1} P(Z_{ikn}(\tau_{ik}) - \frac{1}{2} \bar{J}_n \geq m^{1/2} g(n/m) A/2n \text{ for some } 1 \leq n \leq m) \\ &\rightarrow P(W(t) \geq g(t) \text{ for some } 0 < t \leq 1)(k-1) \\ &= 1 - P^*. \end{aligned}$$

This completes the proof.

We now consider two possible choices of $g(t)$. Let $h = h(k, P^*)$ be a positive constant depending on k and P^* , satisfying

$$(38) \quad (2/\pi)^{1/2} \int_h^\infty \exp \{-\frac{1}{2}x^2\} dx = (1 - P^*)/(k-1).$$

Choose $g(t) = h(k, P^*)$. It is well known (see Billingsley (1968), p. 72) that this choice of $g(t)$ satisfies (35). Also, define $a = a(k, P^*)$ by

$$(39) \quad 1 - \Phi(a) + a\phi(a) + \phi^2(a)/\Phi(a) = (1 - P^*)/(k-1)$$

where Φ and ϕ are the cdf. and density of the standard normal distribution. Values of $a(k, P^*)$ are tabulated in Swanepoel and Geertsema (1976) for various values of k and P^* . Another choice of $g(t)$ is:

$$(40) \quad g(t) = t^{1/2} \alpha^{-1}(\alpha(a) - \log t)$$

where $\alpha^{-1}(x)$ is the inverse function of $\alpha(x) = x^2 + 2\log\Phi(x)$. From Robbins and Siegmund (1970, p. 1413) it follows that $g(t)$ satisfies (35) and the assumptions of theorem 2.2. An explicit expression for $g(t)$ is not possible, but fairly sharp bounds are

$$(41) \quad t^{1/2}[\alpha(a) - \log t]^{1/2} \leq g(t) \leq t^{1/2}[\alpha(a) - \log t + 2\log 2]^{1/2}.$$

If we replace the procedure in (34) by one in which $g(t)$ is taken equal to the upper

bound above, theorem 4.1 would still hold. The right-hand side of (34) then becomes

$$(42) \quad A[a^2 + 2\log\Phi(a) + 2\log 2 + \log(m/n)]^{1/2}/2n^{1/2}.$$

(For other choices of $g(t)$ satisfying (35), see Robbins and Siegmund (1970), p. 1413.)

Remarks

- (a) The procedure proposed, differs substantially from the usual fixed-sample size procedures in the literature (see for example Gupta (1965)). However there is a close correspondence between our approach and well-known truncated sequential tests of hypotheses (see Sen (1974), p. 61).
- (b) If sampling continues up to stage m , the procedure selects a random number of populations as subset. However, if sampling is terminated at stage n , $1 \leq n \leq m-1$, then only *one* population is chosen. This is an attractive feature of the procedure. In fact, if N denotes the number of stages needed for termination and $\ell_N(m)$ the number of populations in the selected subset then $\ell_N(m) \rightarrow 1$ a.s. as $m \rightarrow \infty$, which can be seen as follows:

Suppose sampling is continued up to stage m . From (36) it follows, for all $i \neq j$, that

$$\limsup_{m \rightarrow \infty} |Z_{ijm}(\tau_{ij}) - \bar{J}_m| / (A^2(\log_2 m)/2m)^{1/2} = 1 \text{ a.s.}$$

(This was also proved by Sen and Ghosh (1973).)

Hence, for every $\epsilon > 0$ and infinitely many values of m , we obtain (assuming $\tau_{ij} \geq 0$)

$$\begin{aligned} Z_{ijm}(0) - \bar{J}_m &\geq Z_{ijm}(\tau_{ij}) - \bar{J}_m \\ &\geq (1 - \epsilon)(A^2(\log_2 m)/2m)^{1/2} \\ &\geq (g(1)A)/2m^{1/2}, \end{aligned}$$

so that Π_j will be eliminated by Π_i .

- (c) Since, for all $i \neq j$, $Y_{ij\alpha} - \tau_{ij}$ is symmetrically distributed about 0, it follows from Hájek (1969), p. 105 that the distribution of $Z_{ijn}(\tau_{ij})$ is independent of F . From the proof of theorem 4.1 it therefore follows that for given $\epsilon > 0$ one can specify $m = m(\epsilon)$ independent of F such that $P(\text{CS}) \geq P^* - \epsilon$ for this choice of m . This is

what motivated us to formulate the procedure of this section in terms of rank order statistics.

REFERENCES

- BECHHOFFER, R.E., KIEFER, J. and SOBEL, M. (1968). *Sequential Identification and Ranking Procedures*. Univ. of Chicago Press.
- BILLINGSLEY, P. (1968). *Convergence of Probability Measures*. New York: Wiley.
- CARROLL, R.J. (1975). An asymptotic representation for M-estimators and linear functions of order statistics. *Institute of Statistics Mimeo Series 1006*, University of North Carolina at Chapel Hill.
- CHOW, Y.S. and ROBBINS, H. (1965). On the asymptotic theory of fixed-width sequential confidence intervals for the mean. *Ann. Math. Statist.* **36** 457-462.
- DE WET, T. and VENTER, J.H. (1974). An asymptotic representation of trimmed means with applications. *South African Statist. J.* **8** 127-133.
- GUPTA, S.S. (1965). On some multiple decision rules. *Technometrics* **7** 225-245.
- HAJEK, J. (1969). *Nonparametric statistics*. Holden-Day. San Francisco.
- ITO, K. and McKEAN, H.P. (1965). *Diffusion processes and their sample paths*. Springer.
- JAIN, N.C., JOGDEO, K. and STOUT, W.F. (1975). Upper and lower functions for martingales and mixing processes. *Ann. Probability* **3** 119-145.
- PAULSON, E. (1964). A sequential procedure for selecting the population with the largest mean from k normal populations. *Ann. Math. Statist.* **35** 174-180.
- ROBBINS, H. and SIEGMUND, D.O. (1970). Boundary crossing probabilities for the Wiener process and sample sums. *Ann. Math. Statist.* **41** 1410-1429.
- ROBBINS, H., SOBEL, M. and STARR, N. (1968). A sequential procedure for selecting the largest of k means. *Ann. Math. Statist.* **39** 88-92.
- SEN, P.K. (1974). The invariance principle for one-sample rank order statistics. *Ann. Statist.* **2** 49-62.
- SEN, P.K. and GHOSH, M. (1971). On bounded length sequential confidence intervals based on one-sample rank order statistics. *Ann. Math. Statist.* **42** 189-203.
- SEN, P.K. and GHOSH, M. (1973). A law of the iterated logarithm for one-sample rank order statistics and an application. *Ann. Statist.* **1** 568-576.
- SEN, P.K. and GHOSH, M. (1974). Sequential rank tests for location. *Ann. Statist.* **2** 540-552.
- SWANEPOEL, J.W.H. and GEERTSEMA, J.C. (1973). Nonparametric sequential procedures for selecting the best of k populations. *South African Statist. J.* **7** 85-94.
- SWANEPOEL, J.W.H. and GEERTSEMA, J.C. (1976). Sequential procedures with elimination for selecting the best of k normal populations. *South African Statist. J.* **10** 9-36.
- SWANEPOEL, J.W.H. and Venter, J.H. (1975). On the construction of sequential selection procedures. *South African Statist. J.* **9** 103-118.